The Child Eats the Adult's Supplement

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Abstract

HMTc applies selected infant-food contaminant benchmarks to supplements as voluntary program criteria. Concentration, serving size, preparation and body weight all matter to exposure. The reviewed sources do not establish that all supplement rules assume adults or that adult products are appropriate for children.This briefing distinguishes the original food categories from HMTc's supplement criteria and explains why concentration limits must be read alongside product scope, preparation and intended use.

HMTc applies selected infant-food contaminant benchmarks to supplements as voluntary program criteria. This briefing explains that choice and its limits. A concentration limit is not a complete exposure assessment, and the reviewed sources do not establish that all supplement rules assume an adult body weight or that adult products are appropriate for children.

1 | The dose that matters is per kilogram

For the same contaminant amount swallowed, dividing by a smaller body weight gives a larger dose per kilogram. As an illustration, an identical serving gives a 15-kilogram child about 4.7 times the dose per kilogram received by a 70-kilogram adult. This arithmetic does not establish that children normally consume adult servings or should use adult supplements. Consumption, total dietary exposure and chemical form remain necessary to interpret a measured concentration.

2 | What the reviewed supplement sources establish

USP General Chapter 2232 concerns finished supplement dosage forms labeled as conforming to USP or NF. Its public preview identifies that scope, but does not expose the full numerical table or its derivation [1]. This

briefing therefore does not use the subscription chapter to claim a universal adult-body-weight assumption or to calculate a supplement concentration from an unverified daily limit.

Thailand's Notification 414 (2020) provides an identifiable example: its lead maximum for food supplements is 1 mg/kg, or 1,000 µg/kg, for the product as prepared for consumption [2]. This is a product concentration, not a stated 70-kilogram-adult dose calculation.

ASEAN's 2015 guideline and Singapore HSA's published supplement guidance give lead at 10 ppm. The ASEAN document has an information-only disclaimer about the regulatory agreement at the time; the document alone does not establish current national adoption. HSA also requires safety for the recommended use and target users [3]. A numerical table should therefore not be read as ignoring children or permitting unsafe use.

California OEHHA lists a lead maximum allowable dose level of 0.5 µg/day for Proposition 65. The statutory reproductive-toxicity exemption uses a no-observable-effect criterion at 1,000 times the exposure in question [4]. This daily-exposure safe harbor is not a food concentration limit or a general guarantee of safety for children.

3 | What HMTc holds every supplement to

HMTc's selected program values include lead 10, cadmium 5, inorganic arsenic 10, total mercury 4, nickel 100 and aluminium 500 µg/kg. They must be read with the applicable program standard and product basis; an infant-food benchmark does not become a government limit on a supplement merely because HMTc selects it.

The EU source for cadmium at 5 µg/kg applies to specified liquid formula made from cow-milk proteins or protein hydrolysates; the inorganic-arsenic value of 10 µg/kg also applies to liquid formula. Powder and soy categories have different maxima [5]. The EU nickel value of 100 µg/kg is likewise a liquid-formula maximum, with different powder values [6]. Applying these numbers to tablets or powders does not establish equivalent intake.

The reviewed Australian Schedule 19 and 2024 amendment provide formula lead at 10 µg/kg and aluminium at 500 µg/kg for specified formula. Other aluminium categories include 1,000 µg/kg for soy formula and 200 µg/kg for preterm special-medical products. Preparation rules and the Australia-only transition described by FSANZ matter [7]. The selected 500 µg/kg value is neither the lowest entry in that table nor evidence of a globally unique aluminium rule.

EFSA's nickel opinion establishes a chronic tolerable daily intake of 13 µg/kg body weight. High-percentile exposure exceeded it in many young population groups; mean exposure was generally below or near it [8]. This supports attention to intake and body weight without proving that one supplement concentration assures safety.

4 | The classes where the line is hardest to hold

HMTc applies its supplement criteria across ingredient classes. A product's composition and testing results determine whether it meets the applicable standard. The sources reviewed for this briefing do not quantify compliance rates for shilajit, greens powders or botanical supplements, so no claim about routine failure in those classes is made here.

5 | The same rule, applied twice

The pet-food briefing describes another voluntary use of selected human-food benchmarks. In both cases, HMTc should state the original food category, the product being certified and the basis of comparison. A

benchmark's provenance is different from evidence that it predicts health outcomes in a new population or product.

6 | The tradeoff, stated plainly

Applying infant-food values to all supplement forms can require more stringent product specifications. It remains a program policy choice. A certificate should state the criteria tested and the resulting conformity; it should not imply that an adult product is suitable for children, that every relevant exposure was evaluated, or that a clinical benefit has been established.

References

[1] USP General Chapter : Elemental Contaminants in Dietary Supplements
United States Pharmacopeia. General Chapter <2232>, USP–NF (2022), public publisher preview. Scope: finished dosage forms labeled as conforming to USP or NF. Subscription numerical tables and their derivation were not reviewed.
STANDARD

[2] Thailand MOPH Notification No. 414 (2020) — Standards for Contaminants in Food
Ministry of Public Health, 20 March 2020; Government Gazette, 20 May 2020. Annex 1 lead row: food supplements 1 mg/kg as prepared for consumption. Officially hosted unofficial English translation; Thai text controls. Replaces the former unverified four-jurisdiction grouping.
GOVERNMENT

[3] ASEAN and Singapore health-supplement contaminant guidelines
ASEAN Guidelines on Limits of Contaminants for Health Supplements, version 2.0 (June 2015), and Singapore HSA Health supplements overview (accessed September 2026). Both list lead 10 ppm; legal status and target-user safety requirements require separate attention.
STANDARD

[4] Proposition 65 Safe Harbor Levels — Maximum Allowable Dose Levels
California OEHHA: lead MADL 0.5 µg/day. Daily-exposure safe harbor under Proposition 65; not a food concentration limit or general child-safety guarantee.
GOVERNMENT

[5] Commission Regulation (EU) 2023/915 on maximum levels for certain contaminants in food
European Commission (2023), Annex I. Cadmium 5 µg/kg applies to specified liquid formula made from cow-milk proteins or protein hydrolysates; inorganic arsenic 10 µg/kg applies to liquid formula. Powder and soy categories differ.
REGULATION

[6] Commission Regulation (EU) 2024/1987 amending (EU) 2023/915 as regards maximum levels of nickel in food
European Commission (2024). Nickel maximum 100 µg/kg for liquid formula as marketed; powder maxima differ. These are formula requirements, not legal supplement limits.
REGULATION

[7] Australia New Zealand Food Standards Code, Standard 1.4.1 — Contaminants and Natural Toxicants
FSANZ. Standard 1.4.1; Schedule 19 compilation of 13 September 2024 and amendment F2024L01151. Formula lead 10 µg/kg; aluminium 500 µg/kg for specified formula, 1000 for soy and 200 for preterm special-medical products. Preparation and transition provisions apply.
STANDARD

[8] Update of the Risk Assessment of Nickel in Food and Drinking Water
EFSA CONTAM Panel. EFSA Journal 2020;18(11):6268. Chronic TDI 13 µg Ni/kg body weight/day. High-percentile exposures exceed it in many young groups; mean exposure was generally below or near it. Total-diet assessment, not a supplement concentration limit.
GOVERNMENT

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Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this briefing draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/briefings/supplements-child-protective. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

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